# Smart Queue Management System for Financial Institutions in Rwanda

Milestone 1 (Session 1): Problem Discovery and User Research

Course: Software Engineering and User-Centred Design | Team: T6 | Tag: m1 | Submission date: 07/11/2026

## 1. Project Title and Team Roles

**Project:** Smart Queue Management System (SQMS), a digital solution to reduce waiting time in banks, SACCOs and other financial institutions in Rwanda. A customer requests a queue ticket at the branch, receives an SMS with a queue number and estimated waiting time, and is told by SMS when to go to the counter.

| Team member | Role(s) | Main responsibilities |
|---|---|---|
| Janvier Namahoro | Project Manager / Scrum Master; Business and Requirements Analyst | Planning, Jira board, GitHub releases, meetings, problem statement, SRS, user stories, traceability |
| Jeavis | UX / UI Designer; Technical Lead | Personas, journey maps, prototypes, architecture, competitor technical review |
| Eraste | Lead User Researcher; QA and Documentation Lead; Database Administrator | Interview guide, interviews, affinity mapping, validation, sources.md, quality checks, data model |

Six roles are shared among three team members. Every member must be able to explain and defend the whole project.

## 2. Problem Hypothesis and Problem Statement

### 2.1 Problem hypothesis

Customers of financial institutions in Rwanda lose a lot of time standing in queues because they cannot see their position or know when they will be served. This was tested through our own interviews (Section 6).

### 2.2 Problem statement (course template)

| Template line | Our statement |
|---|---|
| [User group] who [situation] | Customers of banks and SACCOs who visit a branch to deposit, withdraw or apply for a loan |
| struggle to [task/goal] | use their time well while waiting, see their position, estimate their wait, or leave the hall without losing their turn; vulnerable customers have no clear priority route |
| because [root cause] | tickets are manual or missing, the order of service is unclear, and branches record no queue data |
| This causes [consequence] | Waits of about 45 to 90 minutes at peak times (month-end, salary periods, market days); lost time for customers; stress, disputes and interruptions for staff |
| We know this because [evidence] | Five published sources (E1 to E5) and five interviews in Rwamagana (E6): three clients, one teller, one branch manager |

### 2.3 Evidence so far

| # | Evidence | Type | Source |
|---|---|---|---|
| E1 | Members of savings and credit cooperatives said travelling to offices takes long because of queues and called for online and mobile services. | News report | [S1] Rwanda Today, 3 April 2022 |
| E2 | A cooperative member in Musanze said manual operations trigger long queues of customers. | News report | [S5] The New Times |
| E3 | Study at Banque Populaire du Rwanda, Kimironko branch, over ten peak days (start and end of month): about 128 arrivals per hour against a service rate of about 21; utilisation 1.527; almost 45 customers waiting within the first 30 minutes. | Academic queueing study | [S3] Icyemezo et al. (2017) |
| E4 | National Bank of Rwanda mystery shopping (189 visits, 514 scenarios): overall customer-service satisfaction 76.3%. The regulator monitors service quality; high overall satisfaction suggests waiting may be a problem mainly at peak times. | Regulator report | [S4] BNR |
| E5 | In 2019 the Director General of the Rwanda Cooperative Agency said Umurenge SACCO members must go to the cooperative where they opened an account because of limited services. | News report | [S2] The New Times, 29 March 2019 |
| E6 | Five interviews and observation at branches in Rwamagana (5 Nov 2026). | Primary research | Section 6 |

E1 to E5 are secondary evidence, checked on 7 November 2026. Because E4 shows high overall satisfaction, our interviews tested whether the waiting problem is real and concentrated at peak times. Result: all five participants described delays or queue-management difficulty at busy periods, and four reported or estimated waits of 45 to 90 minutes.

### 2.4 Who has this problem?

| Group | How they experience the problem | Role in system |
|---|---|---|
| Customers | Long, unpredictable waits; cannot leave the hall without losing their place | Primary user |
| Priority customers (elderly, pregnant, persons with disabilities) | Stand for long periods; no clear way to be served earlier without asking | Primary user |
| Tellers / customer service officers | Handle crowds, repeated questions and frustrated customers | Primary user |
| Branch managers | Hear complaints but have no recorded data on peak times or delays | Decision-maker / reports user |

## 3. Stakeholder Map (power / interest grid)

Miro board: https://miro.com/app/board/your-board-id (replace with the real board link before submission). PNG export: T6-M1-StakeholderMap.png. The grid below shows the placement of each stakeholder.

| KEEP SATISFIED (high power, lower interest) | MANAGE CLOSELY (high power, high interest) |
|---|---|
| Head office / board<br>National Bank of Rwanda (regulator)<br>Data protection authority (NCSA) | Customers<br>Tellers and customer service officers<br>Branch manager |
| **MONITOR (lower power, lower interest)** | **KEEP INFORMED (lower power, high interest)** |
| Telecom / SMS providers<br>Competing queue-system vendors | Elderly, pregnant and disabled customers and their representatives<br>Security / floor guides<br>IT support team |

| Stakeholder | Role (uses / pays / approves / affected) | Needs and expectations | Strategy |
|---|---|---|---|
| Customers | Use; affected | Short, predictable, fair waits; clear information | Manage closely |
| Tellers / service officers | Use | Clear order of work, fewer disputes, less manual ticketing | Manage closely |
| Branch manager | Uses reports; approves | Live view of queues, staffing data, reports | Manage closely |
| Head office / board | Pays; approves | Efficiency, customer satisfaction, cost control | Keep satisfied |
| Regulator and data protection authority | Approves; oversees | Service quality; protection of personal data such as phone numbers | Keep satisfied |
| Priority customers and representatives | Affected | Priority access and accessibility | Keep informed |
| Security / floor guides | Affected | Orderly lobby | Keep informed |
| IT support | Maintains | Maintainable, documented system | Keep informed |
| SMS providers | Supplies service | Reliable integration | Monitor |

## 4. Competitor Analysis: Three Existing Solutions

Based on public vendor and comparison information [S6 to S9]. Pricing is not confirmed by quotes.

| Criteria | Qmatic | Wavetec | QLess |
|---|---|---|---|
| Origin | Sweden (founded 1981) | UAE (founded 1987) | USA |
| Core offer | Cloud customer-journey and queue management: appointments, check-in, service, feedback | Enterprise queue management for banks: self-check-in, notifications, appointments, dashboards | Virtual queuing and appointment scheduling so customers wait remotely |
| Does well | Strong integrations, multi-location support, real-time notifications | Banking focus, WhatsApp virtual queue, analytics | Simple remote waiting, reminders, walk-ins plus appointments |
| Fails for our context | Custom enterprise pricing; heavy for small institutions | Custom pricing; hardware-oriented | Subscription in USD; no Kinyarwanda or USSD focus |
| Gap we can fill | Low cost; Kinyarwanda and English; SMS for customers without smartphones; priority queues; Rwanda data-protection rules | Same | Same |

**Conclusion:** existing products are mature but priced and designed for large enterprises. A lightweight, locally adapted system that works by SMS, supports Kinyarwanda and has priority queues remains a gap.

## 5. User Research Plan

**Research question:** Does the long, unpredictable waiting problem really exist in financial institutions in Rwanda, and who feels it most? We interviewed five people in person in Rwamagana, with consent. Interviews could be held in Kinyarwanda, English or French; summaries are in English.

| # | Participant type | Where found | What we wanted to learn |
|---|---|---|---|
| P1 | Regular customer (trader or salaried) | SACCO / bank branch, Rwamagana | Visit frequency, wait time, what they do while waiting |

| # | Participant type | Where found | What we wanted to learn |
|---|---|---|---|
| P2 | Priority customer (elderly) | Branch waiting area, Rwamagana | Standing, seating, priority, how they know their turn |
| P3 | Customer | Branch waiting area, Rwamagana | Waiting experience, workarounds |
| P4 | Teller / customer service officer | Branch, Rwamagana | Crowd management, disputes, manual tickets |
| P5 | Branch manager | Branch office, Rwamagana | Complaints, data, staffing and reporting needs |

**Ethics:** each interview began with consent (purpose, about 20 minutes, no right or wrong answers, permission to take notes or record). No names, ID numbers, phone numbers or account details were collected; participants appear only as P1 to P5. Questions asked about past behaviour, not opinions, and the idea was not pitched. Raw notes, audio (with permission) and photos are stored securely.

### Interview guide used

- **Customers:** role and visit frequency; the last time they went to the branch step by step; how long they waited and how they knew their turn; hardest part; what they did while waiting or to avoid it (time or money cost); what would be different if waiting were easier; anything missed; permission to contact again.
- **Staff and managers:** role and a normal busy day; the last time the line was very long; how the next customer is chosen and how tickets or lists are handled; disputes and complaints about waiting; data on busy hours, waiting time and customers served; what would make peak times easier.

## 6. Interview Summaries (anonymised)

All interviews: 5 Nov 2026, in person, interviewer Team T6.

### Interview 1

| | |
|---|---|
| **Participant / role** | P1: 34 years old, businessman; regular customer |
| **Duration / language** | 25 minutes, Kinyarwanda |
| **Context** | SACCO branch in Rwamagana; cash withdrawals and business transactions |
| **Key observations** | Visited at the end of the previous month to withdraw money. Branch was busy; he waited while several customers were served. |
| **Pain points** | Service order unclear: some people who arrived after him were served first. No reliable way to check his position or know when he would be served. |
| **Best quote** | "Some people arrived after me but were served before me. I want customers to be served in a clear and fair order." |
| **Numbers** | Wait about 60 minutes; visits twice a week; no queue fee |
| **Surprise / what changed our thinking** | The issue is not only queue length: customers need to trust that the order of service is fair and transparent. |

## Interview 2

| | |
|---|---|
| **Participant / role** | P2: 66 years old, businesswoman; priority customer (elderly) |
| **Duration / language** | 25 minutes, Kinyarwanda |
| **Context** | Financial institution in Rwamagana; business transactions |
| **Key observations** | Stood for a long time because no seat was available. Came very early to avoid the crowd. |
| **Pain points** | No one called her earlier because of her age; she had to ask a staff member to be served earlier. |
| **Best quote** | "I am old, and I stood for a long time because there was no seat and no one called me earlier." |
| **Numbers** | Wait time and visit frequency not recorded; no queue fee |
| **Surprise / what changed our thinking** | Elderly customers need a clear, respectful way to be served earlier without having to ask staff. |

## Interview 3

| | |
|---|---|
| **Participant / role** | P3: 29 years old, teacher; regular customer |
| **Duration / language** | 20 minutes, Kinyarwanda |
| **Context** | Financial institution in Rwamagana; deposits and occasional withdrawals |
| **Key observations** | Visited in a busy period and found several customers waiting. Unsure how many people were ahead of her. |
| **Pain points** | Could not estimate how long she would wait. Leaving the queue, even briefly, could mean losing her place. The waiting area became crowded. |
| **Best quote** | "When I join the queue, I cannot leave to do another activity because I may lose my place." |
| **Numbers** | Wait about 45 minutes; about three visits a month; no queue fee |
| **Surprise / what changed our thinking** | Customers may value leaving the waiting area and returning when their turn approaches, not only receiving a ticket. |

## Interview 4

| | |
|---|---|
| **Participant / role** | P4: 31 years old, teller / customer service officer; five years of experience |
| **Duration / language** | 25 minutes, Kinyarwanda and English |
| **Context** | Financial institution in Rwamagana; receives customers, checks transaction details, serves at the counter |
| **Key observations** | Higher traffic at month-end and during salary-payment periods. Staff sometimes answer repeated questions about who should be served next. |
| **Pain points** | Manual queue coordination causes confusion about service order. Customers complain about delays; repeated queue questions interrupt service. |
| **Best quote** | "When many customers arrive at the same time, it is difficult to manage the queue manually and concentrate on serving each customer." |
| **Numbers** | Customers may wait 45 to 90 minutes in busy periods; teller volume not recorded |
| **Surprise / what changed our thinking** | The system must support employees as well as customers, with a simple teller screen to call the next ticket. |

### Interview 5

| | |
|---|---|
| **Participant / role** | P5: 45 years old, branch manager; eight years in financial-services management |
| **Duration / language** | 30 minutes, Kinyarwanda and English |
| **Context** | Financial institution in Rwamagana; oversees branch operations, staff coordination and customer service |
| **Key observations** | Higher traffic at month-end, during salary periods and on busy market days. Queue monitoring depends largely on staff observation. |
| **Pain points** | Peak hours are hard to identify without recorded queue data. Complaints about delays are hard to analyse. Staffing relies on experience rather than measured demand. |
| **Best quote** | "A reliable queue system should help us organise customers and provide information that improves service delivery." |
| **Numbers** | Waiting may reach 60 to 90 minutes at peak; daily volume and exact average wait not provided by the branch |
| **Surprise / what changed our thinking** | Management needs reports on customer volume, waiting time and peak periods, not only digital ticketing. |

## 7. Affinity Map

One observation per sticky note in Miro, grouped by similarity; each group became a pain point. Board: https://miro.com/app/board/your-board-id (replace with real link). PNG export: T6-M1-Affinity_Map.png. Every note below is taken from the interview records in Section 6.

| Affinity group | Sticky notes |
|---|---|
| Long and unpredictable waiting | Waited about 60 minutes at month-end (P1) \| Could not estimate the wait or how many people were ahead (P3) \| Cannot leave the queue without losing my place (P3) \| Came very early to avoid the crowd (P2) \| Customers may wait 45 to 90 minutes in busy periods (P4) \| Waiting may reach 60 to 90 minutes at peak (P5) |
| No seating or priority for vulnerable customers | Stood a long time, no seat available (P2) \| No one called me earlier because of my age (P2) \| Had to ask a staff member to be served earlier (P2) |
| Unclear order and disputes | People who arrived after me were served first (P1) \| No reliable way to check my position (P1) \| Manual queue coordination causes confusion about order (P4) \| Repeated questions about who is next (P4) |
| Staff workload and stress | Customers complain about delays (P4) \| Repeated queue questions interrupt service (P4) \| Hard to manage the queue and concentrate on serving (P4) |
| No queue data for management | Peak hours hard to identify without recorded data (P5) \| Complaints about delays are hard to analyse (P5) \| Staffing decisions rely on experience (P5) |

## 8. Five Major Pain Points (each with a quote)

| # | Pain point | Affected | Published evidence | Interview quote |
|---|---|---|---|---|
| PP1 | Long, unpredictable waiting, especially at month-end; cannot leave the queue | Customers, tellers | S1, S3, S5 | "When I join the queue, I cannot leave to do another activity because I may lose my place." (P3, client) |
| PP2 | Standing for long with no seating or priority for vulnerable customers | Priority customers | None found yet | "I am old, and I stood for a long time because there was no seat and no one called me earlier." (P2, client) |
| PP3 | Unclear service order, so customers doubt fairness and tickets cause disputes | Customers, tellers | S3 (arrival pressure) | "Some people arrived after me but were served before me. I want customers to be served in a clear and fair order." (P1, client) |

| # | Pain point | Affected | Published evidence | Interview quote |
|---|---|---|---|---|
| PP4 | Staff workload and interruptions when managing the queue manually | Tellers | None found yet | "When many customers arrive at the same time, it is difficult to manage the queue manually and concentrate on serving each customer." (P4, teller) |
| PP5 | No queue data for management decisions | Branch manager | S3 (peak days measured) | "A reliable queue system should help us organise customers and provide information that improves service delivery." (P5, manager) |

## 9. User Personas (based only on interview data)

Names are illustrative. Devices and connectivity were not recorded in the interviews; entries marked "to confirm" are design assumptions to be validated in Session 2.

### Persona 1: Eric, the busy small-business customer

| | |
|---|---|
| **Profile** | 34 years old, businessman, regular SACCO customer in Rwamagana; visits about twice a week |
| **Devices** | Mobile phone with SMS (to confirm) |
| **Goals** | Finish banking quickly and know when his turn will come |
| **Frustrations** | People who arrived later are served first; no way to check his position; about 60 minutes lost at month-end |
| **Current workaround** | Visits early or at quieter times (to confirm) |
| **Quote** | "Some people arrived after me but were served before me." |
| **Based on** | P1 (with P3 for the leaving-the-queue need) |

### Persona 2: Mama Rose, the priority customer

| | |
|---|---|
| **Profile** | 66 years old, businesswoman; elderly priority customer |
| **Devices** | Basic phone, SMS only (to confirm) |
| **Goals** | Be served early and respectfully without having to ask |
| **Frustrations** | No seat; stood a long time; no one called her earlier |
| **Current workaround** | Comes very early; asks a staff member to serve her earlier |
| **Quote** | "I am old, and I stood for a long time because there was no seat and no one called me earlier." |
| **Based on** | P2 |

### Persona 3: Grace, the teacher on a tight schedule

| | |
|---|---|
| **Profile** | 29 years old, teacher; deposits money and makes occasional withdrawals about three times a month |
| **Devices** | Mobile phone with SMS (to confirm) |
| **Goals** | Know how long she will wait and be able to leave and return |
| **Frustrations** | Cannot estimate the wait; fears losing her place; crowded waiting area; about 45 minutes waiting |
| **Current workaround** | Stays in the queue (no alternative reported) |
| **Quote** | "When I join the queue, I cannot leave to do another activity because I may lose my place." |
| **Based on** | P3 |

### Persona 4: Claudine, the teller

| | |
|---|---|
| **Profile** | 31 years old, teller / customer service officer, five years of experience |
| **Devices** | Counter banking system (to confirm) |
| **Goals** | Serve customers in a clear, fair order and concentrate on each transaction |
| **Frustrations** | Month-end and salary-period crowds; repeated questions about who is next; complaints about delays |
| **Current workaround** | Manual queue coordination |
| **Quote** | "It is difficult to manage the queue manually and concentrate on serving each customer." |
| **Based on** | P4 |

### Persona 5: Jean Paul, the branch manager

| | |
|---|---|
| **Profile** | 45 years old, branch manager, eight years in financial-services management |
| **Devices** | Laptop and phone (to confirm) |
| **Goals** | Keep service smooth, plan staffing from measured demand, improve service delivery |
| **Frustrations** | No recorded queue data; complaints hard to analyse; staffing relies on experience |
| **Current workaround** | Staff observation and coordination |
| **Quote** | "A reliable queue system should help us organise customers and provide information that improves service delivery." |
| **Based on** | P5 |

## 10. Proposed Solution Direction

A customer arrives at the branch and enters a mobile phone number on a smart queue device. The system automatically gives the next sequential queue number and sends an SMS with the number and the estimated waiting time. When the turn approaches, another SMS tells the customer to go to the counter, so customers can run errands and return shortly before their turn. Priority queue numbers go to pregnant women, elderly people and persons with disabilities according to the institution's policy, tellers get a simple screen to call the next ticket, and the system records queue data so managers can see busy periods. The application is not designed in this milestone; design starts in later milestones based on the validated pain points.

## 11. Milestone 1 Evidence Checklist

| Required deliverable | Where | Status |
|---|---|---|
| Project title and team roles | Section 1 | Done |
| Problem statement with evidence | Section 2 | Done |
| Stakeholder map (Miro link + PNG) | Section 3; T6-M1-StakeholderMap.png | Add real Miro link and PNG |
| Competitor analysis (3 solutions) | Section 4 | Done |
| 3+ anonymised interview summaries | Sections 5 and 6 (5 interviews) | Done |
| Affinity map (Miro link + PNG) | Section 7; T6-M1-Affinity_Map.png | Add real Miro link and PNG |
| 5 pain points, each with a quote | Section 8 | Done |
| Personas (1 to 2 required; 5 provided) | Section 9 | Done |
| Proposed solution direction | Section 10 | Done |
| sources.md and ai-use-log.md updated | docs/research/ | Update in repository |
| GitHub release tag m1 + link on class platform | Repository | Pending |

**Approval question:** can we show that this problem exists using evidence rather than assumptions? Yes: published sources E1 to E5 and five interviews (E6) agree that waits are long and unpredictable at peak times, that order of service is unclear, and that branches hold no queue data.

## Sources


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